

MIDTERM LAB TASK 2

USING FUNCTION

Using Functions

Problem 1.

Create an $n \times n$ Multiplication table using **Nested FOR Loop**. The user must enter the number of rows and columns that will be displayed in the Table.

Sample Output 1

```
How many rows:10
How many cols:10
      Multiplication Table

  1   2   3   4   5   6   7   8   9  10
  2   4   6   8  10  12  14  16  18  20
  3   6   9  12  15  18  21  24  27  30
  4   8  12  16  20  24  28  32  36  40
  5  10  15  20  25  30  35  40  45  50
  6  12  18  24  30  36  42  48  54  60
  7  14  21  28  35  42  49  56  63  70
  8  16  24  32  40  48  56  64  72  80
  9  18  27  36  45  54  63  72  81  90
 10  20  30  40  50  60  70  80  90 100
```

Sample Output 2.

```
How many rows:3
How many cols:5
      Multiplication Table

  1   2   3   4   5
  2   4   6   8  10
  3   6   9  12  15
```

SOURCE CODE

```
1 rows = int(input("How many rows:"))
2 columns = int(input("How many cols:"))
3 print("Multiplication Table")
4
5 for a in range(1,rows +1):
6     for b in range(1, columns +1):
7         print(f"{a * b:5d}", end = " ")
8     print("\n")
```

OUTPUT

```
How many rows:10
How many cols:10
Multiplication Table
 1    2    3    4    5    6    7    8    9   10
 2    4    6    8   10   12   14   16   18   20
 3    6    9   12   15   18   21   24   27   30
 4    8   12   16   20   24   28   32   36   40
 5   10   15   20   25   30   35   40   45   50
 6   12   18   24   30   36   42   48   54   60
 7   14   21   28   35   42   49   56   63   70
 8   16   24   32   40   48   56   64   72   80
 9   18   27   36   45   54   63   72   81   90
10   20   30   40   50   60   70   80   90  100
```

Problem 2. Create a bank program that will allow the user to perform the ff: Use Functions as necessary

```
*****
      ABCCDE ATM
*****
1.Show Balance
2.Deposit
3.Withdraw
4.Exit
*****
```

```
Enter your choice (1-4): 1
*****
Your balance is $0.00
*****
```

```
*****
Enter your choice (1-4): 2
*****
Enter an amount to be deposited: 1000
*****
```

```
Enter your choice (1-4): 1
*****
Your balance is $1000.00
*****
```

```
Enter your choice (1-4): 3
*****
Enter amount to be withdrawn: 250
*****
```

```
*****
Enter your choice (1-4): 1
*****
Your balance is $750.00
*****
```

SOURCE CODE

```
1 def show_balance(balance):
2     print(f"Your balance is PHP{balance:.2f}")
3
4 def deposit(balance):
5     amount = float(input("Enter amount to deposit: "))
6     if amount > 0:
7         balance += amount
8         print(f"Successfully deposited PHP{amount:.2f}")
9     else:
10        print("Invalid amount.")
11    return balance
12
13 def withdraw(balance):
14     amount = float(input("Enter amount to withdraw: "))
15     if amount > 0 and amount <= balance:
16         balance -= amount
17         print(f"Successfully withdrew PHP{amount:.2f}")
18     elif amount > balance:
19         print("Insufficient funds!")
20     else:
21         print("Invalid amount.")
22     return balance
23
24 def main():
25     atm_name = "Cash Madali"
26     balance = 0.0
27     choice = 0
28
29     print("=====")
30     print(f"        {atm_name}        ")
31     print("=====\\n")
32     while choice != 4:
33         print("\\n=====")
34         print("\\nMenu:")
```

```
4         print("\\nMenu:")
5         print("1. Show Balance")
6         print("2. Deposit")
7         print("3. Withdraw")
8         print("4. Exit")
9         print("\\n=====")
10
11     try:
12         choice = int(input("\\nEnter your choice (1-4): "))
13     except ValueError:
14         print("Please enter a valid number (1-4).")
15         continue
16
17     if choice == 1:
18         show_balance(balance)
19     elif choice == 2:
20         balance = deposit(balance)
21     elif choice == 3:
22         balance = withdraw(balance)
23     elif choice == 4:
24         print(f"Thank you for using {atm_name}.")
25     else:
26         print("Invalid choice. Please try again.")
27
28 main()
```

OUTPUT

```
=====
Cash Madali
=====

Menu:
1. Show Balance
2. Deposit
3. Withdraw
4. Exit
=====

Enter your choice (1-4):
```

```
=====
Enter your choice (1-4): 1
Your balance is PHP0.00
=====

Menu:
1. Show Balance
2. Deposit
3. Withdraw
4. Exit
=====

Enter your choice (1-4):
```

```
=====
Enter your choice (1-4): 2
Enter amount to deposit: 500
Successfully deposited PHP500.00
=====

Menu:
1. Show Balance
2. Deposit
3. Withdraw
4. Exit
=====

Enter your choice (1-4):
```

```
=====
Enter your choice (1-4): 3
Enter amount to withdraw: 500
Insufficient funds!
=====

Menu:
1. Show Balance
2. Deposit
3. Withdraw
4. Exit
=====

Enter your choice (1-4):
```

```
=====
Menu:
1. Show Balance
2. Deposit
3. Withdraw
4. Exit
=====

Enter your choice (1-4): 4
Thank you for using Cash Madali.

...Program finished with exit code 0
Press ENTER to exit console.
```